

PropHedge

Hedging Prop Firm Challenges with Static Drawdown:
Hedge Ratios, Path-Dependent PnL and Expected Value

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Abstract

Proprietary trading firms (prop firms) sell access to evaluation accounts for a one-time fee. Profits on these accounts are partially paid out once the evaluation is passed, while losses are capped at the fee. This asymmetric payoff structure can be exploited by opening an opposing, smaller position on a real brokerage account (live account). We formalise this approach for challenges with a *static* drawdown as a two-stage random walk with absorbing barriers, derive the PnL of every path and the expected value in closed form, and optimise the hedge ratios with respect to the Sortino ratio of the outcomes. Our main findings are: (i) for suitable hedge ratios, all terminal outcomes of the challenge become positive; (ii) within the practically relevant range, the expected value grows approximately linearly with the maximum drawdown; (iii) the rules of a challenge have a considerably larger impact on its expected value than its price; (iv) assuming a win rate of 50 %, daily drawdown limits and consistency rules can both be reduced to a common upper bound on position size. We further quantify the robustness of the strategy against denied payouts.

Keywords: prop firm, evaluation account, hedging, hedge ratio, gambler’s ruin, Sortino ratio, expected value, static drawdown

JEL Classification: G11, G13, C61

1 Introduction

1.1 What Are Prop Firms?

Traditional proprietary trading firms trade with their own capital and give their traders a share of the profits. In recent years, a second business model aimed at retail traders has emerged: for a one-time fee, a trader purchases a *challenge* (also called an *evaluation*) on a simulated account with a nominal account size, for example \$50,000. If the trader reaches a predefined profit target without breaching a loss limit, they receive a so-called *funded account*. On this account, profits above a certain threshold are actually paid out at a fixed percentage (profit split).

1.2 How Do Challenges Work?

A typical challenge is characterised by the following rules:

Profit target Percentage gain relative to the account size that must be reached to pass the evaluation.

Maximum drawdown Loss limit that terminates the account immediately once reached. It can be defined as *static* (fixed relative to the starting capital) or *trailing* (moving relative to the account’s high-water mark).

Daily drawdown Maximum permitted loss within a single trading day.

Consistency rule No single day may contribute more than a certain share of the total profit.

Payout conditions Profit threshold, profit split and minimum number of trading days required for a payout on the funded account.

The providers' business model relies on the fact that the vast majority of participants fail the evaluation and that the fees exceed the payouts to successful traders. For the trader, this results in a highly asymmetric payoff: the loss is capped at the fee, while the gain equals the funded account profit multiplied by the profit split. Structurally, this resembles a long option position whose premium is the challenge fee.

1.3 Contribution

We study how this asymmetry can be exploited systematically by hedging the challenge with an opposing position on a live account. The idea originates from the *Prop Arbitrage Simulator* by ATJ Research [1], which simulates such a hedging strategy across several phases. This paper translates the approach into a closed-form analytical model that determines the net PnL for every possible path through the challenge, derives the expected value and risk metrics, and optimises the hedge ratios. We explicitly note that hedging between a prop account and a personal account violates the terms of service of almost all providers. This is modelled explicitly as the risk of denied payouts in Section 6.2.

2 Hedging Mechanics

2.1 Basic Principle

Every trade on the prop account is simultaneously opened on a live account in the same instrument, but in the *opposite direction*. If the trader buys a contract on the prop account, they sell on the live account, and vice versa. The position on the live account is always *smaller* than the one on the prop account.

Definition 1 (Hedge ratio). Let X denote the position size on the prop account. The position size on the live account is

$$X^{\text{live}} = -h \cdot X, \quad h \in [0, 1].$$

The parameter h is called the *hedge ratio*. We write h_E for the evaluation phase and h_F for the funded phase.

If the prop account moves by an amount Δ , the live account moves by $-h\Delta$ (before fees). If the trader fails the challenge, the live account therefore gains a fraction h of the drawdown, fully or partially compensating the fee. If the trader passes the challenge, the live account loses a fraction h of the profit target, which is meant to be more than offset by the payout in the funded phase.

2.2 Why $h < 1$?

A full hedge ($h = 1$) would neutralise every gain on the prop account with an equally large loss on the live account. Since the prop account carries no capital of its own, however, the two sides are not symmetric: the loss on the prop account is capped at the fee, while its gain is paid out only after passing and only partially. The optimal hedge ratio balances this asymmetry and therefore lies in the interior of the interval. In addition, the live account ties up real capital (Section 5.4).

2.3 Exploiting the Payoff Structure

Without a hedge, the challenge is a lottery: with high probability the fee is lost, with low probability a large payout follows. With a hedge, the losing case is turned into a small gain, while the winning case is reduced by the hedging costs. The hedge thus reshapes the distribution of outcomes. With a suitable choice of (h_E, h_F) , all terminal outcomes can even become positive.

3 Scope: Static Drawdown

This paper is restricted exclusively to challenges with a *static* drawdown, where the loss limit is fixed at $A(1 - D)$ regardless of the account's history.

With a *trailing* drawdown, the loss limit moves up with the account's high-water mark. The remaining distance to the loss limit, and hence the probability of failure, then depends on the entire path so far. A constant hedge ratio is no longer optimal in this case; it would have to be recomputed after every trade depending on the current profit, and the live position would have to be adjusted accordingly.

Such dynamic hedging is problematic in practice for several reasons:

- **Execution risk:** Every adjustment requires additional orders. Bad fills, slippage and spreads accumulate and distort the effective hedge ratio.
- **Asynchrony:** The prop account and the live account are executed on different platforms. Time lags between fills create unintended net exposure.
- **Granularity:** Contracts can only be traded in whole units (e.g. Micro E-mini S&P 500, MES). A continuously varying hedge ratio can therefore only be approximated, especially for small account sizes.

With a static drawdown, the geometry of the problem remains unchanged throughout each phase, so that one constant hedge ratio per phase is sufficient.

4 Model

4.1 Assumptions

1. Each trade wins or loses the fixed amount rA with probability $\frac{1}{2}$, where r is the fraction of the account size risked per trade.
2. The drawdown is static.
3. No other rule (daily drawdown, consistency, minimum trading days) is breached. Section 6.4 shows that this can be ensured by an upper bound on r .
4. The live position is always executed exactly at ratio h (no slippage).
5. The starting capital of the funded account is $A_F = A(1 + T_E)$; a fraction φ of the funded profit is paid out. In the numerical example, $\varphi = 1$.

4.2 Notation

4.3 Transition Probabilities

Under Assumption 1, the account balance follows a symmetric random walk with step size rA that is absorbed at the upper barrier $+TA$ (pass) or the lower barrier $-DA$ (fail). By the

Symbol	Meaning	Numerical example
A	Account size	\$50,000
C	Challenge price	\$149
r	Fraction risked per trade	2 %
T_E, D_E	Profit target / max. drawdown, evaluation	8 % / 4 %
T_F, D_F	Payout target / max. drawdown, funded	8 % / 4 %
f_E	Fee per trade on the prop account	\$5
f_L	Fee per trade on the live account	\$2
φ	Profit split	100 %
h_E, h_F	Hedge ratio, evaluation / funded	optimised

Table 1: Model parameters and values used in the numerical example.

classical gambler's ruin result [2, 4],

$$p_E = \frac{D_E}{D_E + T_E}, \quad p_F = \frac{D_F}{D_F + T_F}, \quad (1)$$

where p_E denotes the probability of passing the evaluation and p_F the conditional probability of reaching the payout target in the funded phase. Notably, these probabilities depend only on the ratio T/D and *not* on the position size r .

This yields three terminal outcomes with unconditional probabilities

$$P_{FE} = 1 - p_E \quad (\text{evaluation failed}), \quad (2)$$

$$P_{PO} = p_E p_F \quad (\text{payout reached}), \quad (3)$$

$$P_{FF} = p_E (1 - p_F) \quad (\text{funded account lost}). \quad (4)$$

The probabilities are thus fully determined by the profit target and the maximum drawdown of the respective challenge; there is no universal standard case. For the parameters of the numerical example in Table 1 ($T = 8\%$, $D = 4\%$ in both phases), we obtain $p_E = p_F = \frac{1}{3}$, i.e. $P_{FE} = 66.7\%$, $P_{PO} = 11.1\%$ and $P_{FF} = 22.2\%$. A challenge with $T = 10\%$ and $D = 5\%$ leads to the same result, whereas $T = 6\%$ and $D = 4\%$ already yields $p = 40\%$.

4.4 Expected Number of Trades and Fees

For a symmetric random walk absorbed at $+a$ and $-b$ steps, the expected duration is $a \cdot b$ [2]. With $a = T/r$ and $b = D/r$, it follows that

$$n_E = \frac{T_E D_E}{r^2}, \quad n_F = \frac{T_F D_F}{r^2}. \quad (5)$$

The fees per phase are

$$F_E = n_E (f_E + f_L \mathbf{1}_{\{h_E > 0\}}), \quad F_F = n_F f_L. \quad (6)$$

Since the funded phase is only reached with probability p_E , the expected total fees are

$$\mathbb{E}[F] = F_E + p_E F_F. \quad (7)$$

5 Path PnL and Expected Value

5.1 Composition of the PnL

The net PnL of a path consists of four components: the challenge fee C , the PnL of the live hedge, the payout of the funded account, and the trading fees. If a phase ends at the drawdown,

the live account gains hD ; if it ends at the target, it loses hT , each relative to the capital of that phase.

Evaluation passed (intermediate state).

$$\Pi_{PE} = -C - h_E T_E A - F_E. \quad (8)$$

Evaluation failed.

$$\Pi_{FE} = -C + h_E D_E A - F_E. \quad (9)$$

Funded account lost.

$$\Pi_{FF} = \Pi_{PE} + h_F D_F A_F - F_F. \quad (10)$$

Payout reached.

$$\Pi_{PO} = \Pi_{PE} - h_F T_F A_F + \varphi T_F A_F - F_F. \quad (11)$$

Figure 1 shows the paths as a decision tree, evaluated at the Sortino-optimal hedge ratios of the numerical example, $h_E = 0.25$ and $h_F = 0.65$.

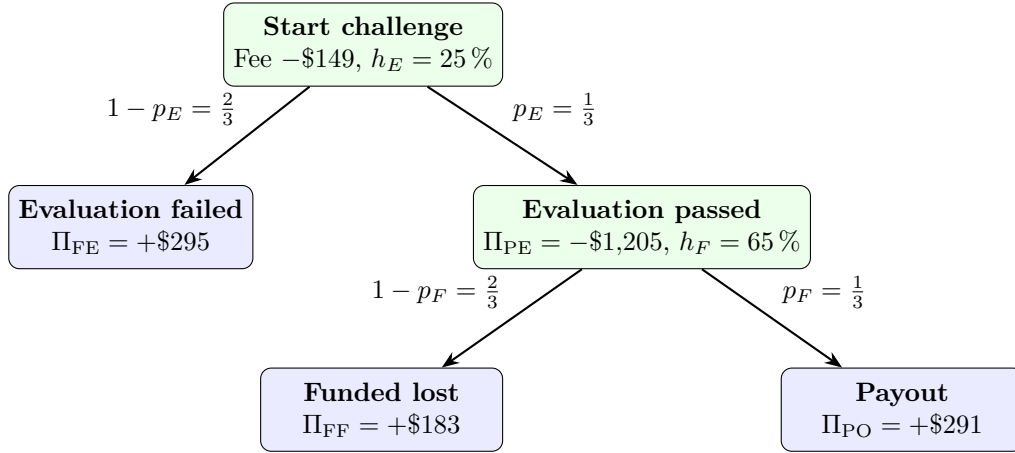


Figure 1: Decision tree of the hedged challenge in the numerical example with $h_E = 0.25$, $h_F = 0.65$. Green: intermediate states, blue: terminal states.

5.2 Expected Value

The expected value is the probability-weighted sum of the terminal outcomes:

$$\mathbb{E}[\Pi] = P_{FE} \Pi_{FE} + P_{PO} \Pi_{PO} + P_{FF} \Pi_{FF}. \quad (12)$$

In the numerical example, this gives

$$\mathbb{E}[\Pi] = \frac{2}{3} \cdot 295 + \frac{1}{9} \cdot 291 + \frac{2}{9} \cdot 183 \approx \$269.67.$$

5.3 Risk Measure and Optimisation

The hedge ratios are chosen according to a risk measure. We use the Sortino ratio [3]

$$S(h_E, h_F) = \frac{\mathbb{E}[\Pi]}{\sigma_-}, \quad \sigma_-^2 = \sum_{i \in \{FE, PO, FF\}} P_i \min(\Pi_i - \mathbb{E}[\Pi], 0)^2. \quad (13)$$

Unlike its usual application, the Sortino ratio here does not refer to a time series of returns but to the distribution of the three terminal outcomes Π_{FE} , Π_{PO} and Π_{FF} . The downside deviation σ_- measures how far the terminal outcomes fall below the expected value, weighted by their probabilities. Maximising S relates the expected value to this downside deviation: we seek the hedge ratios for which the outcomes of the individual paths lie as close together as possible, i.e. with as little volatility across outcomes as possible. Ideally, every path leads to nearly the same PnL regardless of how the challenge ends.

The optimisation is carried out by grid search over $h_E, h_F \in \{0, 0.05, \dots, 1\}$. Figure 2 shows the result for the numerical example. The Sortino ratio is sharply concentrated around the optimum at $h_E = 0.25$ and $h_F = 0.65$, where it reaches a value of approximately 6.6. At this point, all three terminal outcomes lie between \$183 and \$295 (Figure 1). Even small deviations from the optimum, particularly in the evaluation hedge, cause the Sortino ratio to drop significantly. An h_E that is too high makes passing the evaluation so expensive that the funded phase can no longer compensate for the loss; an h_E that is too low leaves the challenge fee uncovered in the losing case.

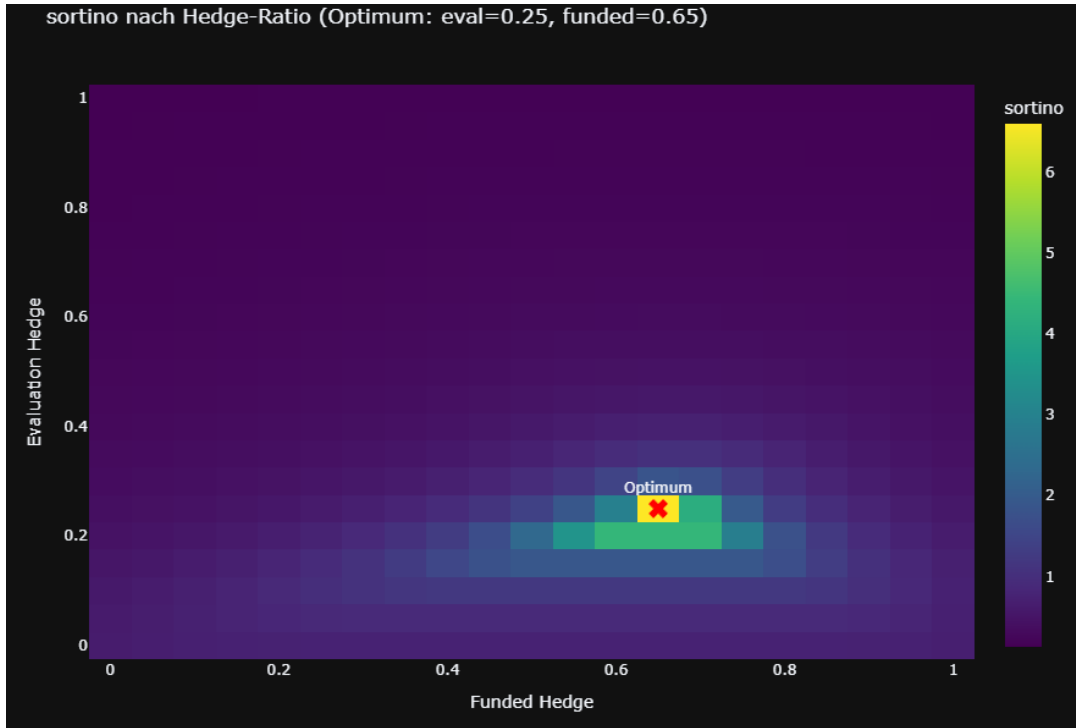


Figure 2: Sortino ratio of the terminal outcomes as a function of the hedge ratios h_E (vertical) and h_F (horizontal) for the numerical example. The optimum lies at $h_E = 0.25$, $h_F = 0.65$.

5.4 Capital Requirement

The prop account is simulated, whereas the live account carries real capital. The worst running PnL occurs on the payout path immediately before the payout, when both hedge losses have been realised but the payout has not yet been received:

$$\Pi_{\min} = \Pi_{PE} - h_F T_F A_F - F_F. \quad (14)$$

In the numerical example, $\Pi_{\min} = -\$4,029$. Together with the margin for the live position, this value determines the minimum capital that must be held.

6 Results

6.1 The Expected Value Scales with the Maximum Drawdown

Table 2 shows the expected value according to (12) with $h_E = 0.25$ and $h_F = 0.65$ for different maximum drawdowns, all other parameters unchanged. If the drawdown is increased equally in both phases ($D_E = D_F = D$), the expected value grows approximately linearly by about \$130–\$140 per percentage point in the range from 2 % to 8 %.

D	p_E	$\mathbb{E}[\Pi], D_E = D_F = D$	$\mathbb{E}[\Pi], \text{ only } D_E = D$
2 %	20.0 %	−\$5.8	\$107.8
3 %	27.3 %	\$127.0	\$197.4
4 %	33.3 %	\$269.7	\$269.7
5 %	38.5 %	\$412.4	\$328.7
6 %	42.9 %	\$550.2	\$377.3
7 %	46.7 %	\$680.7	\$417.5
8 %	50.0 %	\$803.0	\$451.0

Table 2: Expected value as a function of the maximum drawdown (other parameters as in the numerical example).

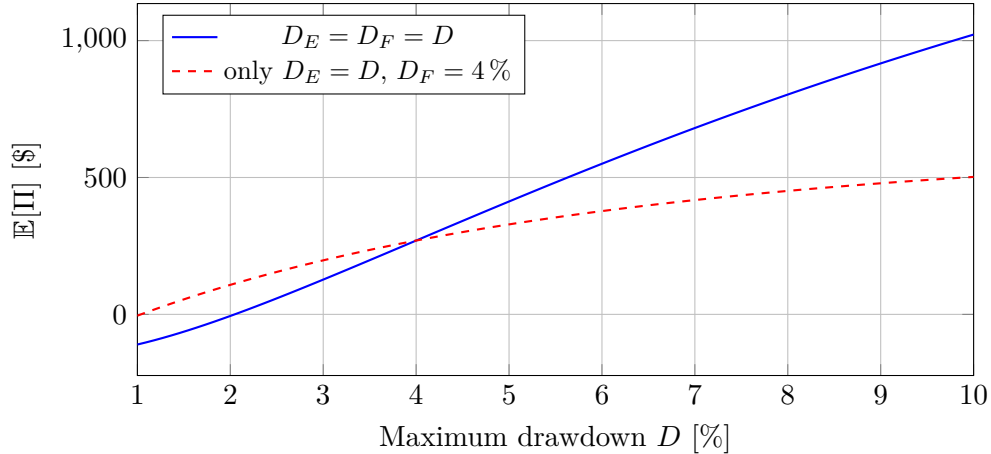


Figure 3: Expected value as a function of the maximum drawdown according to Equation (12) with $h_E = 0.25$, $h_F = 0.65$.

A larger drawdown increases the pass probabilities p_E and p_F and thus the expected payout. For hedging, a larger drawdown means more room: in the losing case the live account gains hDA , so that the challenge fee can be fully recovered even with smaller hedge ratios. This is counteracted by the trading fees, which grow linearly with D according to (5). Strictly speaking, $\mathbb{E}[\Pi]$ is not linear in D ; however, the quadratic dependence on $p_E p_F$ is offset by the saturation of $D/(D + T)$ in such a way that an almost linear relationship emerges within the practically relevant range (Figure 3). If only the evaluation drawdown is increased, the gain flattens considerably.

6.2 Robustness Against Denied Payouts

Since most prop firms prohibit hedging, one has to expect that a payout may be denied. Let $q \in [0, 1]$ be the probability that a reached payout is actually paid. The terminal state “payout”

then splits: with probability q the outcome is Π_{PO} , with probability $1 - q$ it is

$$\Pi_{\text{PO}}^{\text{denied}} = \Pi_{\text{PE}} - h_F T_F A_F - F_F = \Pi_{\text{min}},$$

since the hedge loss of the funded phase has already been realised when the payout is rejected. The two variants of the payout state differ by exactly the payout $\varphi T_F A_F$. Hence

$$\mathbb{E}[\Pi](q) = \mathbb{E}[\Pi] - (1 - q) p_E p_F \varphi T_F A_F, \quad (15)$$

where $\mathbb{E}[\Pi]$ is the expected value from (12). The breakeven payout rate is

$$q^* = 1 - \frac{\mathbb{E}[\Pi]}{p_E p_F \varphi T_F A_F}. \quad (16)$$

In the numerical example with $h_E = 0.25$ and $h_F = 0.65$, we obtain $q^* = 1 - 269.67/480 \approx 43.8\%$. The strategy therefore remains profitable as long as fewer than about 56% of the reached payouts are denied.

It is nevertheless worth noting that, at the optimal hedge, by far the most likely terminal state, failing the evaluation ($P_{\text{FE}} = \frac{2}{3}$), is already profitable at +\$295 without any payout. The risk of the strategy is thus concentrated on the payout path.

6.3 Rules Dominate Price

Since the fee C enters every terminal outcome with a factor of -1 , $\partial \mathbb{E}[\Pi]/\partial C = -1$: every dollar of challenge fee reduces the expected value by exactly one dollar. The rules, in contrast, act multiplicatively through the pass probabilities on the comparatively large payout $\varphi T_F A_F$. Table 3 compares both effects.

Change relative to numerical example	$\mathbb{E}[\Pi]$	$\Delta \mathbb{E}[\Pi]$
Numerical example	\$269.7	–
Price C : \$149 $\rightarrow$ \$99	\$319.7	+\$50.0
Price C : \$149 $\rightarrow$ \$299	\$119.7	–\$150.0
Drawdown $D_E = D_F$: 4% $\rightarrow$ 5%	\$412.4	+\$142.7
Drawdown $D_E = D_F$: 4% $\rightarrow$ 3%	\$127.0	–\$142.7
Profit target T_E : 8% $\rightarrow$ 6%	\$367.9	+\$98.2
Profit target T_E : 8% $\rightarrow$ 10%	\$195.5	–\$74.2
Profit split φ : 100% $\rightarrow$ 80%	\$173.7	–\$96.0

Table 3: Sensitivity of the expected value to price and rules.

In the numerical example, a single additional percentage point of drawdown is worth almost as much as the entire challenge fee. When selecting a challenge, the rules (drawdown, profit target, profit split) should therefore be compared first and the price only afterwards. A more expensive challenge with a more generous drawdown is generally superior to a cheap one with a tight drawdown.

6.4 Daily Drawdown and Consistency Rule as a Position Size Bound

According to (1), with a win rate of exactly 50% the pass probabilities do not depend on the position size r . The position size is therefore a free parameter that can be used to comply with further rules. If at most k trades are executed per day:

- **Daily drawdown** with limit d : the maximum daily loss is krA . The rule is satisfied if $kr \leq d$.

- **Consistency rule** with share c : the maximum daily profit is krA and may not exceed cTA . The rule is satisfied if $kr \leq cT$.

Both rules thus reduce to the same type of constraint, namely an upper bound on the position size:

$$r \leq r_{\max} = \frac{1}{k} \min(d, cT). \quad (17)$$

Within this bound, the rules do not affect the distribution of terminal outcomes. Their only costs arise through the number of trades, which grows with $1/r^2$ according to (5), and through the longer duration of the challenge. Table 4 illustrates this effect.

r	n_E	n_F	$\mathbb{E}[\Pi]$
0.5 %	128	128	−\$650.3
1 %	32	32	\$85.7
2 %	8	8	\$269.7
4 %	2	2	\$315.7

Table 4: Effect of position size on the number of trades and the expected value (numerical example, $h_E > 0$).

A daily drawdown or consistency rule is consequently costly exactly when it lowers $r_{\max}$ so far that the fees consume the payout term. For the evaluation of a challenge, both rules can therefore be compared uniformly via their implied bound $r_{\max}$.

For practical implementation, it follows that r should be chosen as large as permitted, i.e. $r = r_{\max}$. Since the expected number of trades decreases with $1/r^2$ according to (5), a maximal r minimises both the transaction costs on the prop account and the hedging costs on the live account. Because $r_{\max} = \min(d, cT)/k$, this requires keeping the number of trades per day k as small as possible, ideally $k = 1$. Few but larger trades per day are thus the most cost-efficient strategy under both rules.

7 Limitations

- A win rate of exactly 50 % and symmetric gains and losses are an idealisation; spread and slippage shift the live hedge slightly into negative expected value.
- The result of Section 6.2 shows that the prohibition of hedging is the central risk. The rate q is unknown and provider-dependent.
- The assumption $A_F = A(1 + T_E)$ deviates from providers that reset the funded account to the original account size.
- Only a single payout is considered. Multiple consecutive payouts on the same funded account would increase the expected value.
- Contract granularity and margin requirements are only approximately accounted for.

8 Conclusion

For challenges with a static drawdown, the hedged prop firm challenge can be described fully in closed form. The expected value follows from the PnL of the three possible paths, weighted by their probabilities. The hedge reshapes the distribution such that, for suitable hedge ratios, all terminal outcomes become positive. Within the relevant range, the expected value grows

approximately linearly with the maximum drawdown, and the rules of a challenge matter considerably more for its profitability than its price. With a win rate of 50 %, daily drawdown and consistency rules can be treated uniformly as an upper bound on position size. The largest risk remains the denial of payouts: in the numerical example, at least about 44 % of reached payouts must actually be paid for the strategy to be profitable.

Disclaimer

This paper is a purely analytical study. It does not constitute investment, financial or legal advice and is not an invitation to violate the terms of service of prop firms or brokers. Hedging between a prop account and a personal account is prohibited by most providers and may result in account termination and denied payouts. Trading futures and other financial instruments involves substantial risk of loss.

Code Availability

The Python code for computing the path PnL, the expected value and the Sortino-optimal hedge ratios is publicly available at <https://github.com/Lennart967/PropHedge>.

Declaration of Generative AI Use

Claude (Anthropic) was used in the preparation of this paper to draft the text based on the model and code developed by the author, to produce the LaTeX typesetting and figures, and to verify calculations. The idea, modelling, implementation and interpretation of the results are the author's own; the author has reviewed all content and takes full responsibility for the paper.

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